

WSI Viewer Release Cheat Sheet

Development -> Staging -> Production rehearsal -> Production -> Tag

Environments

Environment	Port	Runtime / identity	Data boundary
Development	8081	Live source; red banner	Deidentified development images and annotations
Staging	8082	Candidate JAR; yellow banner	Deidentified staging images and annotations
Rehearsal	8083	Exact staging JAR; production mode; loopback only	Deidentified production-marked images; rehearsal annotations
Production	8080	Frozen validated JAR; no banner	Authorized clinical images and production annotations

Normal release

1. Development gate

```
cd /Users/dm026/Downloads/wsi-server_works
git status --short
./mvnw clean test
node --test src/test/js/*.test.js
git push origin feature/multichannel-viewer
```

STOP: Git must be clean and every test must pass.

2. Build and install staging

```
wsi-release stage --step
wsi-release verify staging
```

Browser 8082: staging banner; expected deidentified slides; pan, zoom, channels, annotations and exports; no CSRF/403 errors.

3. Rehearse production mode

```
wsi-release rehearse --step
wsi-release verify rehearsal
```

Browser 8083: no banner; production layout; exact staging behavior; rehearsal-only data. Port 8080 remains available.

4. Promote exact rehearsed candidate

```
wsi-release promote --step
wsi-release verify production
```

Type PROMOTE only after staging and rehearsal match and the rollback backup verifies.

5. Tag after production validation

```
wsi-release tag production-YYYY-MM-DD-description
```

Browser 8080: normal production layout; authorized slides; pan, zoom, channels, annotations and exports; no CSRF/403 errors; all other environments remain isolated.

Troubleshooting and Recovery

Status and verification

```
wsj production status
wsj staging status
wsj rehearsal status
wsj development status
wsj-release status
wsj-release verify staging
wsj-release verify rehearsal
wsj-release verify production
```

Logs

```
wsj production logs
wsj staging logs
wsj rehearsal logs
wsj development logs
```

Press Control-C to stop following a log.

Rollback

```
wsj-release history
wsj-release rollback --step
wsj-release verify production
```

Rollback restores the prior JAR, metadata, checksum and configuration. It never restores or overwrites annotations.

Troubleshooting modes

```
wsj-release stage --dry-run
wsj-release stage --step --verbose
wsj-release rehearse --dry-run
wsj-release promote --dry-run
wsj-release history
```

--dry-run: plan only. --step: Enter runs, p repeats, q exits safely. Never bypass PROMOTE or ROLLBACK.

Stop conditions

- Dirty Git tree or failed tests
- Git HEAD differs from staging commit
- Recorded and calculated SHA-256 differ
- Staging and rehearsal differ
- Environment marker, identity, root or port mismatch
- Expected port does not bind
- Browser regression in security, layout, images, annotations or exports

Do not repair a mismatch by editing build metadata or checksum files.

Marker rule

Identity	Required marker
development	.wsj-environment-development
staging	.wsj-environment-staging
production / rehearsal	.wsj-environment-production

Privacy: Never copy production slides into development, staging or rehearsal. Verify deidentification of filenames, metadata, embedded labels, thumbnails, macro images and associated files.

Exactly one environment marker must exist. A missing, multiple or cross-environment marker prevents startup before the server accepts requests.